# Physiology Test Bank

# Quiz (2) MCQs

## Section (1) Cardiovascular Physiology

### 1- (Cardiac Cycle)

**Q1. Closure of the atrioventricular valves marks the beginning of:**

A. Atrial systole
B. Rapid ventricular filling
C. Ventricular diastole
D. Ventricular systole

Ans: D

**Q2. The second heart sound (S2) is primarily produced by:**

A. Closure of the semilunar valves
B. Rapid ventricular filling
C. Opening of the atrioventricular valves
D. Closure of the atrioventricular valves

Ans: A

**Q3. During the isovolumetric contraction phase of the cardiac cycle, which of the following best describes the change in ventricular pressure?**

A. Ventricular pressure increases gradually while ventricular volume decreases.
B. Ventricular pressure increases rapidly while ventricular volume remains constant.
C. Ventricular pressure decreases rapidly while ventricular volume remains constant.
D. Ventricular pressure remains unchanged while ventricular volume increases rapidly.

Ans: B

**Q4. A patient's ventricular pressure rises above aortic pressure during systole. Which immediate mechanical event should occur?**

A. Mitral valve opens and ventricular filling begins
B. Tricuspid valve opens
C. Aortic valve opens and ventricular ejection begins
D. Aortic valve closes and isovolumetric relaxation begins

Ans: C

### 2- (Normal ECG)

**Q5. Which ECG component represents ventricular depolarization?**

A. T wave
B. QRS complex
C. P wave
D. PR interval

Ans: B

**Q6. What does the PR interval primarily represent?**

A. The time required for complete ventricular repolarization
B. The time from ventricular depolarization to ventricular repolarization
C. The time required for complete ventricular depolarization
D. The time from atrial depolarization to ventricular depolarization

Ans: D

**Q7. An ECG shows a regular rhythm with 4 large squares between consecutive R waves. Assuming standard ECG paper speed, what is the approximate heart rate?**

A. 100 beats/min
B. 75 beats/min
C. 150 beats/min
D. 50 beats/min

Ans: B

**Q8. A patient has a markedly prolonged QT interval on the ECG. Which physiological process is most directly prolonged?**

A. Atrial contraction and ventricular filling
B. Ventricular depolarization and repolarization
C. AV nodal conduction and atrial repolarization
D. Atrial depolarization and conduction

Ans: B

## 3- (Abnormal ECG and Cardiac Arrhythmias)

**Q9. Which feature is characteristic of atrial fibrillation?**

A. Complete absence of ventricular activity
B. Prolonged PR interval with progressively dropped QRS complexes
C. Absence of organized P waves with an irregularly irregular ventricular rhythm
D. Regular rhythm with identical P waves before every QRS

Ans: C

**Q10. Which ECG finding is most characteristic of a first-degree AV block?**

A. Absent T waves
B. Prolonged PR interval followed by a QRS complex
C. Shortened PR interval
D. Widened QRS without P waves

Ans: B

**Q11. A 58-year-old man with chronic kidney disease presents with muscle weakness. Laboratory investigations show a markedly elevated serum potassium concentration. An ECG is performed to assess the cardiac effects of electrolyte abnormality. Which ECG finding would most likely be observed?**

A. Increased amplitude of all P waves
B. Tall, peaked T waves
C. Shortened QRS complexes with prominent P waves
D. Persistent ST-segment elevation as the earliest typical finding

Ans: B

**Q12. An ECG shows ST-segment elevation in leads corresponding to a specific region of the heart in a patient with acute chest pain. What does this finding most strongly suggest?**

A. First-degree AV block
B. Acute myocardial injury/ischemia involving that region
C. Sinus bradycardia
D. Normal ventricular repolarization

Ans: B

## 4- (Physiology of Vascular Endothelium)

**Q13. Which substance produced by vascular endothelial cells is an important physiological vasodilator?**

A. Thromboxane A2
B. Nitric oxide
C. Angiotensin II
D. Endothelin-1

Ans: B

**Q14. A healthy blood vessel is exposed to normal blood flow without any vascular injury. Which endothelial response is most important for preventing unnecessary thrombus formation?**

A. Secretion of factors that promote coagulation.
B. Release of mediators that supress fibrinolysis.
C. Release of mediators that reduce platelet adhesion.
D. Expression of surface proteins that enhance thrombin generation.

Ans: C

**Q15. A 55-year-old man has had poorly controlled hypertension for several years. Recent investigations suggest impaired endothelial function in his systemic arteries. Which endothelial change would most likely contribute to increased vascular tone in this patient?**

A. Increased endothelial antithrombotic activity
B. Increased inhibition of smooth muscle contraction
C. Reduced nitric oxide-mediated vasodilation
D. Increased nitric oxide bioavailability

Ans: C

**Q16. A 67-year-old patient with poorly controlled type 2 diabetes presents with signs of early diabetic retinopathy, including microaneurysms and retinal edema. Which of the following endothelial-related mechanisms most likely contributes to the development of these retinal findings?**

A. Increased nitric oxide release causing formation of microaneurysms.
B. Decrease in local proinflamatory markers.
C. Increase vascular permeability and abnormal angiogenesis.
D. Decreased endothelin-1 production causing impaired vasoconstriction.

Ans: C

## Section (2): Respiratory Physiology

**Q1. Which structure belongs primarily to the respiratory zone?**

A. Primary bronchi
B. Alveoli
C. Trachea
D. Terminal bronchioles

Ans: B

**Q2. Which pressure normally falls below atmospheric pressure during inspiration and rises above atmospheric pressure during expiration?**

A. Pulmonary arterial pressure
B. Transpulmonary pressure
C. Intrapleural pressure
D. Alveolar pressure

Ans: D

**Q3. A 24-year-old man is brought to the emergency department after sustaining a penetrating injury to the right side of his chest. On examination, breath sounds are markedly reduced on the right side, and a chest X-ray shows a right-sided pneumothorax. Which pressure change most likely caused the collapse of the affected lung?**

A. Increased atmospheric pressure
B. Loss of the normal negative intrapleural pressure
C. Increased intra-alveolar pressure
D. More negative intrapleural pressure

Ans: B

**Q4. A healthy person takes a deep inspiration. Which combination of changes occurs?**

A. Diaphragm contracts, thoracic volume increases, alveolar pressure decreases
B. External intercostals relax, thoracic volume increases, alveolar pressure increases
C. Diaphragm relaxes, thoracic volume decreases, alveolar pressure increases
D. Diaphragm contracts, thoracic volume decreases, alveolar pressure increases

Ans: A

**Q5. Lung compliance is best defined as:**

A. The tendency of the lungs to return to their resting volume after inflation
B. The resistance encountered by air as it moves through the respiratory tract
C. The ease with which the lungs expand in response to a change in pressure
D. The rate at which air moves through the conducting airways

Ans: C

**Q6. Which statement best describes the direct effect of pulmonary surfactant?**

A. It increases the rate of gas diffusion.
B. It reduces the alveolar surface tension.
C. It enhances lung recoil during expiration.
D. It decreases lung compliance.

Ans: B

**Q7. A 62-year-old woman with progressive shortness of breath is diagnosed with pulmonary fibrosis. Pulmonary function tests were performed. Which combination of changes would most likely be present?**

A. Increased compliance and increased lung volumes
B. Normal compliance with markedly increased airway resistance
C. Decreased compliance and increased work of breathing
D. Increased compliance and reduced elastic recoil

Ans: C

**Q8. A male infant is born prematurely at 29 weeks of gestation and develops rapid breathing and increasing respiratory distress shortly after birth. Which pathological event is most likely the cause of the distress?**

A. Increased tendency of alveoli to collapse
B. Decreased work of breathing
C. Increased lung compliance
D. Decreased alveolar surface tension

Ans: A

**Q9. According to Fick's law of diffusion, gas diffusion across the respiratory membrane increases when:**

A. The partial-pressure gradient increases
B. The diffusion coefficient decreases
C. Surface area decreases
D. Membrane thickness increases

Ans: A

**Q10. In a healthy upright person, pulmonary blood flow is greatest in which region of the lung?**

A. Middle region only
B. Base
C. Apex
D. It is equal throughout the lung

Ans: B

**Q11. As part of regulating pulmonary blood flow, hypoxic pulmonary vasoconstriction (HPV) due to local alveolar collapse will result in:**

A. Decrease perfusion to well-ventilated segment.
B. Decrease ventilation/ perfusion (V/Q) ratio.
C. Decrease perfusion to non-ventilated segment.
D. Decrease total pulmonary blood flow.

Ans: C

**Q12. In an upright person, an alveolus at the lung apex has relatively low blood flow but receives ventilation. Which consequence is most likely?**

A. Complete absence of ventilation
B. Relatively high V/Q ratio
C. Very low V/Q ratio
D. V/Q ratio approaches zero

Ans: B

**Q13. The oxygen-haemoglobin (O2-Hb) dissociation curve demonstrate that:**

A. Hb binds irreversibly to oxygen when O_{2} partial pressure is low.
B. Hb releases more O_{2} when the O_{2} partial pressure is lower than 40 mmHg.
C. Hb releases more O_{2} when the O_{2} partial pressure is higher than 90 mmHg.
D. Hb binds irreversibly to oxygen when O_{2} partial pressure is high.

Ans: B

**Q14. The oxygen-hemoglobin dissociation curve describes the relationship between two important respiratory parameters. What are these two parameters?**

A. Alveolar oxygen concentration and percent hemoglobin saturation.
B. Arterial oxygen concentration and hemoglobin concentration.
C. Arterial oxygen concentration and percent of hemoglobin saturation.
D. Alveolar oxygen concentration and Arterial oxygen concentration.

Ans: C

**Q15. Why is the adult hemoglobin (HbA) molecule considered an efficient oxygen carrier?**

A. Because of the higher affinity of oxygen binding to hemoglobin compared to other gasses.
B. Because oxygen is not soluble in plasma.
C. Because the affinity of hemoglobin binding to oxygen depends on oxygen concentration.
D. Because of irreversible binding of oxygen to hemoglobin.

Ans: C

**Q16. During vigorous exercise, skeletal muscle temperature and CO_{2} production increase. What happens to the oxyhemoglobin dissociation curve?**

A. Shifts to the right, increasing oxygen affinity
B. Shifts to the left, facilitating oxygen delivery
C. Shifts to the right, facilitating oxygen delivery
D. Shifts to the left, increasing oxygen affinity

Ans: C

**Q17. The largest proportion of CO_{2} in blood is transported in the form of:**

A. Carbonic acid
B. Bicarbonate ions
C. Carbaminohemoglobin
D. Dissolved CO_{2}

Ans: B

**Q18. What mechanism aims to maintain a normal electrical gradient across the RBC's membrane?**

A. The Haldane's effect.
B. The hydrogen shift.
C. The chloride shift.
D. The Bohr's effect.

Ans: C

**Q19. During vigorous exercise, a skeletal muscle consumes more O_{2} and produces increased amounts of CO_{2} and H^{+} How does this change affect hemoglobin's interaction with oxygen in the active muscle?**

A. Hemoglobin has a higher affinity for O_{2}, facilitating greater O_{2} loading
B. Hemoglobin has a lower affinity for O_{2}, facilitating greater O_{2} unloading
C. Hemoglobin has a higher affinity for O_{2}, reducing O_{2} delivery
D. Hemoglobin has a lower affinity for O_{2}, reducing O_{2} unloading

Ans: B

**Q20. As blood reaches the pulmonary capillaries, oxygen binds to hemoglobin. How does this facilitate CO_{2} unloading?**

A. Oxygen binding prevents bicarbonate conversion
B. Oxygen binding causes chloride to remain permanently inside RBCs
C. Oxygen binding promotes release of CO_{2} and H from hemoglobin
D. Oxygen binding increases hemoglobin's ability to bind CO_{2}

Ans: C

**Q21. Which region of the brainstem contains the major respiratory centers responsible for generating the basic respiratory rhythm?**

A. Hypothalamus
B. Spinal cord only
C. Medulla
D. Cerebellum

Ans: C

**Q22. Voluntary control of breathing is primarily mediated by:**

A. Spinal autonomic ganglia
B. Cerebral cortex
C. Pulmonary capillaries
D. Medulla only

Ans: B

**Q23. A person voluntarily holds their breath. After a period of time, the increasing urge to breathe becomes difficult to suppress. Which change contributes most strongly to this response?**

A. Increased arterial O_{2} stimulating pulmonary stretch receptors
B. Increased arterial CO_{2} and H concentration stimulating chemoreceptors
C. Increase plasma $H^+$ which stimulates Hering-Breuer reflex.
D. Decreased arterial O_{2} which stimulate central chemoreceptors.

Ans: B

**Q24. A 45-year-old patient is being assessed after receiving a medication that slows cellular metabolism for colon cancer. During assessment, the patient's arterial blood gas shows an increase in arterial CO_{2} (PaCO_{2}). Which of the following immediate physiological responses would most likely occur as a compensation by the respiratory centers?**

A. Suppression of respiratory muscle activity
B. A decrease in heart rate and respiratory rate
C. An increase in the rate and depth of breathing
D. A decrease in the sensitivity of the respiratory center to CO_{2}

Ans: C